

CAPSTONE PROJECT REPORT
FINAL REPORT
SUPPLY CHAIN MANAGEMENT

By Samhita Bhat

Context

1.	Understanding the business problem	3
	1.1 Defining the problem	3
	1.2 Need for the project	3
	1.3 Understanding business/social opportunity	3
2.	Exploratory Data Analysis	3
	2.1 Data Collection and background	3
	2.2 Data overview	4
	2.3 Univariate analysis	4
	2.4 Bivariate analysis	7
	2.5 Multivariate analysis	13
3.	Data Preprocessing	14
	3.1 Duplicate value check	14
	3.2 Data cleaning	14
	3.3 Anomalous value check	14
	3.4 Missing value check	14
	3.5 Feature Engineering	14
	3.6 Outlier Check and Treatment	15
	3.7 Class Imbalance handling	15
	3.8 Data Preparation for Modelling	15
	3.9 Feature Scaling	15
	3.10 Data Leakage Handling	15
4.	Model Building - Baseline model	15
	4.1 Metric of Choice	15
	4.2 Baseline Model Development	16
	4.3 Model Performance Evaluation	16
5	Model Building – Advanced Models	17
	5.1 Models Used	18
	5.2 Model Performance Comparison	18
6	Model Performance Improvement using Hyperparameter Tuning	20
	6.1 Top 3 models for Tuning	20
	6.2 Performance of Tuned Models	20
7	Model Performance Comparison and Final Model Selection	21
	7.1 Comparison of tuned models (Test data)	21
	7.2 Final model Selection	21
8	Actionable Insights and Recommendations	22
	8.1 Key Insights	22
	8.2 Actionable Business Recommendations	22
	Appendix A	23
	Appendix B	25
	Appendix C	37
	Appendix D	37

Figures

Sl.No	Title	Pg.No
-------	-------	-------

1	Product_wg_ton	4
2	Statistical summary (Numerical variables)	5
3	Statistical summary (Categorical variables)	6
4	Correlation heatmap	7
5	Numerical vs Target	8
6	Categorical vs Target	10
7	Binary columns	12
8	Pairplot	13
9	A1 Missing Value Check	23
10	A3 Outlier detection	24
11	'WH_capacity_size'	25
12	zone	25
13	WH_regional_zone	26
14	Num_refill_req_l3m	26
15	Transport_issue_l1y	27
16	Competitor_in_mkt	28
17	Retail_shop_num	28
18	Wh_owner_type	29
19	Distributor_num	29
20	Flood_impacted	30
21	Flood_proof	31
22	Electric_supply	31
23	Dist_from_hub	32
24	Workers_num	32
25	Wh_est_year	33
26	Storage_issue_reported_l3m	34
27	Temp_reg_mach	34
28	Approved_wh_govt_certificate	35
29	Wh_breakdown_l3m	35
30	Govt_check_l3m	36
31	Observed vs Predicted plot	37

Tables

Sl.No	Title	Pg.No
1	Data categories	4
2	Missing value treatment	14
3	LR (model 1)	16
4	LR (model 2)	17
5	LR (model 1 vs model 2)	17
6	Training performance Comparison	18
7	Testing performance Comparison	18
8	Tuned model performance	20
9	Comparison of tuned models	21
10	Comparison of tuned and untuned models (training)	37
11	Comparison of tuned and untuned models (testing)	37

1. Understanding the business Problem

1.1 Defining the problem

A fast-moving consumer goods (FMCG) company that entered the instant noodles market two years ago is facing challenges in balancing demand and supply across its warehouses nationwide. In some regions, demand far exceeds supply, while in others, excess inventory is leading to wastage and higher costs. This mismatch results in significant inventory losses and missed sales opportunities. To remain competitive and profitable in the highly dynamic FMCG sector, the company needs a data-driven approach to optimize its supply chain and align production with regional demand patterns.

The company wants to use historical supply chain data to determine the optimum quantity of product to ship to each warehouse. Solving this problem will enable the company to minimize inventory costs, reduce stockouts, and drive higher sales through better demand forecasting and efficient supply chain management.

1.2 Need for the project

This is an important project as inaccurate demand planning can cause overstocking, understocking, high operational costs, reduced customer satisfaction, inventory wastage, delay in deliveries and revenue loss due to missed sales opportunity.

To address these issues, predictive analytics can identify weak points like transport issues, geographic issues and help forecast future product demand, improve delivery efficiency, reduce wastage and supply chain costs

1.3 Understanding business/social opportunity

This project has several business/social benefits and opportunities:

Business Opportunity:

- 2 Improved inventory distribution leading to fewer shortages.
- 3 Efficient warehouse utilization.
- 4 Lower operational costs
- 5 Increased profitability and revenue generation.
- 6 Efficient logistics and transportation planning.

Social opportunities:

- 7 Reduce in product wastage
- 8 Faster deliveries leading to improved customer satisfaction
- 9 More sustainable supply chain operations.
- 10 Faster product availability in urban and rural areas

2. Exploratory Data Analysis

2.1 Data collection and background

The dataset contains historical supply chain data of an FMCG company operating in the instant noodles market. It includes information related to warehouse characteristics, transportation

issues, competitors, geographic locations, storage conditions, workforce and government certifications.

Data categories:

Variable categories	Variables
Unique ID	Ware_house_ID, WH_Manager_ID
Warehouse Characteristics	Location_type, WH_capacity_size,
Geographic Zones	Zone, WH_regional_zone
Operational Variables	num_refill_req_l3m, transport_issue_l1y, Competitor_in_mkt, wh_owner_type, distributor_num, workers_num,
Infrastructure variables	flood_proof, electric_supply, temp_reg_mach
Logistic Variables	dist_from_hub,
Target variable	product_wg_ton
Others	flood_impacted, wh_est_year, approved_wh_govt_certificate, wh_breakdown_l3m, storage_issue_reported_l3m, govt_check_l3m

Table 1. Data categories

2.2 Data overview

- Total number of rows: 25,000
- Total number of columns: 24
- Target variable: product_wg_ton
- Problem type: Regression problem

2.3 Univariate analysis

Target variable - product_wg_ton

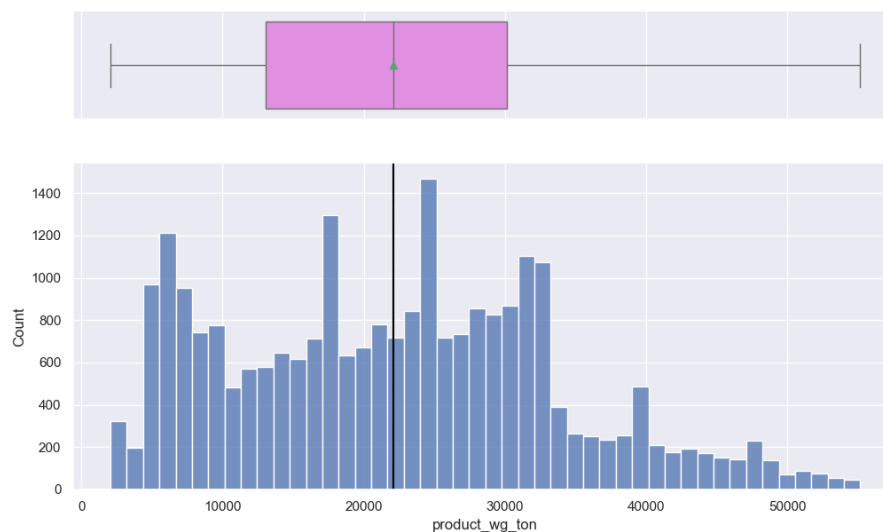


Fig 1. Product_wg_ton

Observations:

- The distribution positively skewed and ranges from 2,000 to 55,000 tons.
- The mean and median are almost similar which indicates moderate balance in the distribution.
- Most of the product weight fall between 13,000 to 30,000.
- The boxplot does not show any outliers.

Insights:

- Product movement varies significantly across warehouses, indicating diverse operational capacities ad regional demand patterns.
- Warehouse operations are not uniformly distributed across the network.

Statistical summary

	count	mean	std	min	25%	50%	75%	max
num_refill_req_l3m	25000.0	4.089040	2.606612	0.0	2.0	4.0	6.0	8.0
transport_issue_l1y	25000.0	0.773680	1.199449	0.0	0.0	0.0	1.0	5.0
Competitor_in_mkt	25000.0	3.104200	1.141663	0.0	2.0	3.0	4.0	12.0
retail_shop_num	25000.0	4985.711560	1052.825252	1821.0	4313.0	4859.0	5500.0	11008.0
distributor_num	25000.0	42.418120	16.064329	15.0	29.0	42.0	56.0	70.0
flood_impacted	25000.0	0.098160	0.297537	0.0	0.0	0.0	0.0	1.0
flood_proof	25000.0	0.054640	0.227281	0.0	0.0	0.0	0.0	1.0
electric_supply	25000.0	0.656880	0.474761	0.0	0.0	1.0	1.0	1.0
dist_from_hub	25000.0	163.537320	62.718609	55.0	109.0	164.0	218.0	271.0
workers_num	24010.0	28.944398	7.872534	10.0	24.0	28.0	33.0	98.0
wh_est_year	13119.0	2009.383185	7.528230	1996.0	2003.0	2009.0	2016.0	2023.0
storage_issue_reported_l3m	25000.0	17.130440	9.161108	0.0	10.0	18.0	24.0	39.0
temp_reg_mach	25000.0	0.303280	0.459684	0.0	0.0	0.0	1.0	1.0
wh_breakdown_l3m	25000.0	3.482040	1.690335	0.0	2.0	3.0	5.0	6.0
govt_check_l3m	25000.0	18.812280	8.632382	1.0	11.0	21.0	26.0	32.0
product_wg_ton	25000.0	22102.632920	11607.755077	2065.0	13059.0	22101.0	30103.0	55151.0

Fig 2. Statistical summary (Numerical variables)

Observations:

- Warehouses serve an average of 4,985 retail shops and work with approximately 42 distributors, suggesting extensive networks.
- The average distance from the hub is 163.53km, with distances ranging from 55km to 271km, indicating varying levels of logistics accessibility.
- Warehouses employ an average of 29 workers, although workforce varies across warehouses.
- The average number of storage issue reported in the last 3 months is 17 times, indicating the storage management is a common operational challenge.
- Government inspections frequently occur with an average 18 inspections in the last 3 months.

- The target variable 'product_wg_ton' has an average value of 22,102.63 tons, with values ranging from 2,065 to 55,151 tons, indicating variation in warehouse product movement.

	count	unique	top	freq
Ware_house_ID	25000	25000	WH_100000	1
WH_Manager_ID	25000	25000	EID_50000	1
Location_type	25000	2	Rural	22957
WH_capacity_size	25000	3	Large	10169
zone	25000	4	North	10278
WH_regional_zone	25000	6	Zone 6	8339
wh_owner_type	25000	2	Company Owned	13578
approved_wh_govt_certificate	24092	5	C	5501

Fig 3. Statistical summary (Categorical variables)

Observations:

- Majority of the warehouses are located in rural areas (22957) indicating a strong rural presence within the supply chain network.
- Large capacity warehouses are the most common category, suggesting a focus on handling high product volumes.
- The North zones contain the highest number of warehouses, making it one of the most operationally significant regions.
- Zone 6 is the most represented regional zone.
- Most of the warehouses are company owned, indicating that the organisations prefer direct ownership over rental arrangements.
- Among warehouses with available certification information, Grade C certification is the most common government approval category.

Insights:

- The supply chain network is heavily concentrated in rural regions, indicating that these locations play a critical role in product distribution.
- Warehouses differ considerably in terms of capacity, workforce size, and product movement, suggesting varying operational requirements across facilities.
- Storage management and infrastructure-related factors are likely to influence warehouse performance due to the relatively high number of reported storage issues.
- The large variation in product weight handled across warehouses supports the need for predictive models to identify factors driving warehouse productivity and operational efficiency.

2.4 Bivariate analysis

Correlation heatmap

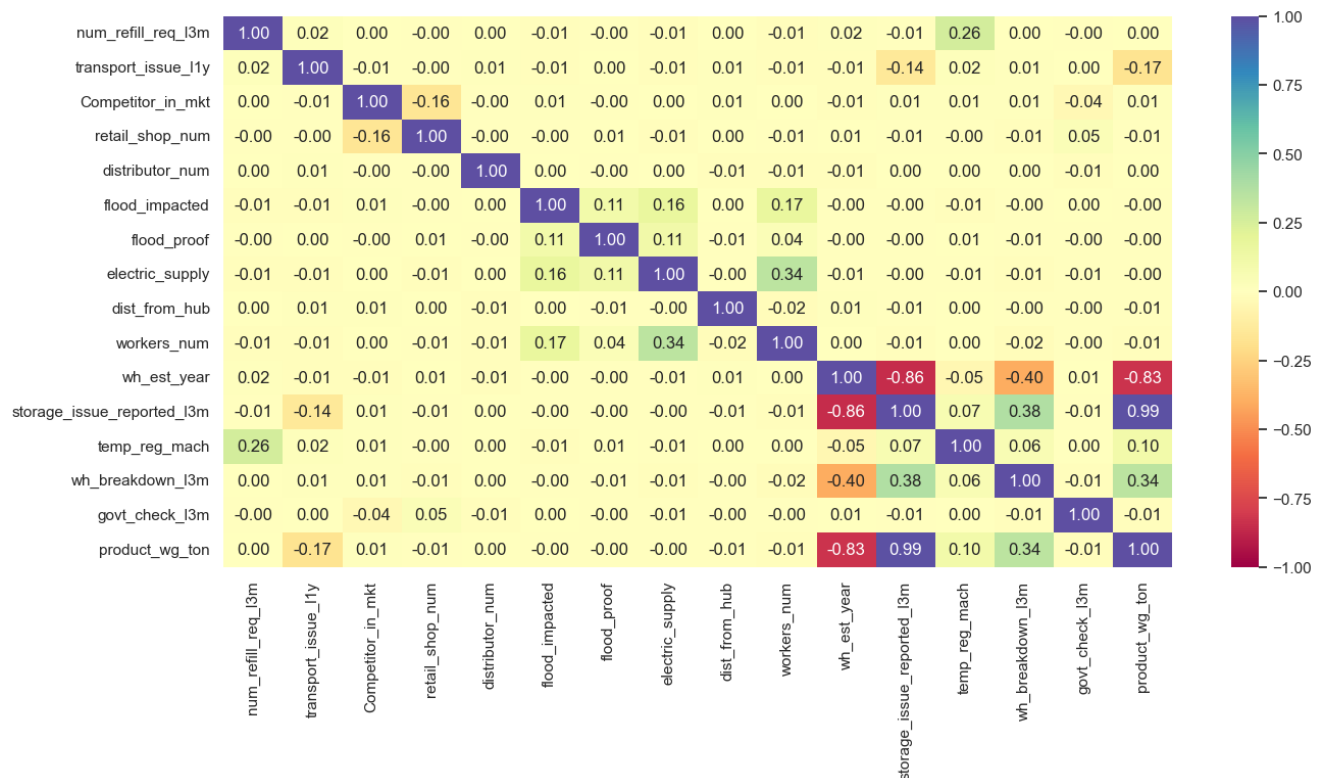


Fig 4. Correlation heatmap

Observations:

- **product_wg_ton & storage_issue_reported_l3m (0.99)** - Extremely strong positive correlation. Warehouses reporting more storage problems are associated with higher shipment quantities.
- **product_wg_ton & wh_est_year (-0.83)** - Strong Negative Correlation. Newer warehouses tend to ship lower quantities.
- **product_wg_ton & wh_breakdown_l3m (0.34)** - Moderative positive correlation. Warehouses with higher shipment quantities tend to have more operational breakdowns.
- **product_wg_ton & temp_reg_mach (0.10)** - Weak positive correlation. Warehouses with temperature regulation machinery may handle larger shipment quantities.
- **product_wg_ton & transport_issue_l1y (-0.17)** - Weak negative correlation. Transportation issues may slightly reduce shipment efficiency.
- **wh_breakdown_l3m & wh_est_year (-0.40)** - Moderative negative correlation. Newer warehouses have less operational breakdowns.

- **wh_breakdown_l3m & storage_issue_reported_l3m (0.38)** - Moderative positive correlation. Warehouses with more storage issues tend to face more operational breakdowns.
- **temp_reg_mach & num_refill_req_l3m (0.26)** - Warehouses with temperature regulation machinery might require more refills of products.
- **storage_issue_reported_l3m & wh_est_year (-0.86)** - Strong negative correlation. Older warehouses tend to report more storage related issues.
- **workers_num & flood_impacted (0.17)** - Weak negative correlation. Flood affected warehouses may require more workforce for maintenance and operations.
- **workers_num & electric_supply (0.34)** - Warehouses with electric supply tend to have more workers.
- **electric_supply & flood_impacted (0.16)** - Weak positive correlation. Warehouses that are impacted by flood may have backup electric systems.
- **Competitor_in_mkt & retail_shop_num (-0.16)** - Weak negative correlation. Regions with many competitors may have fewer retail shops selling this company's product.

Numerical vs Target

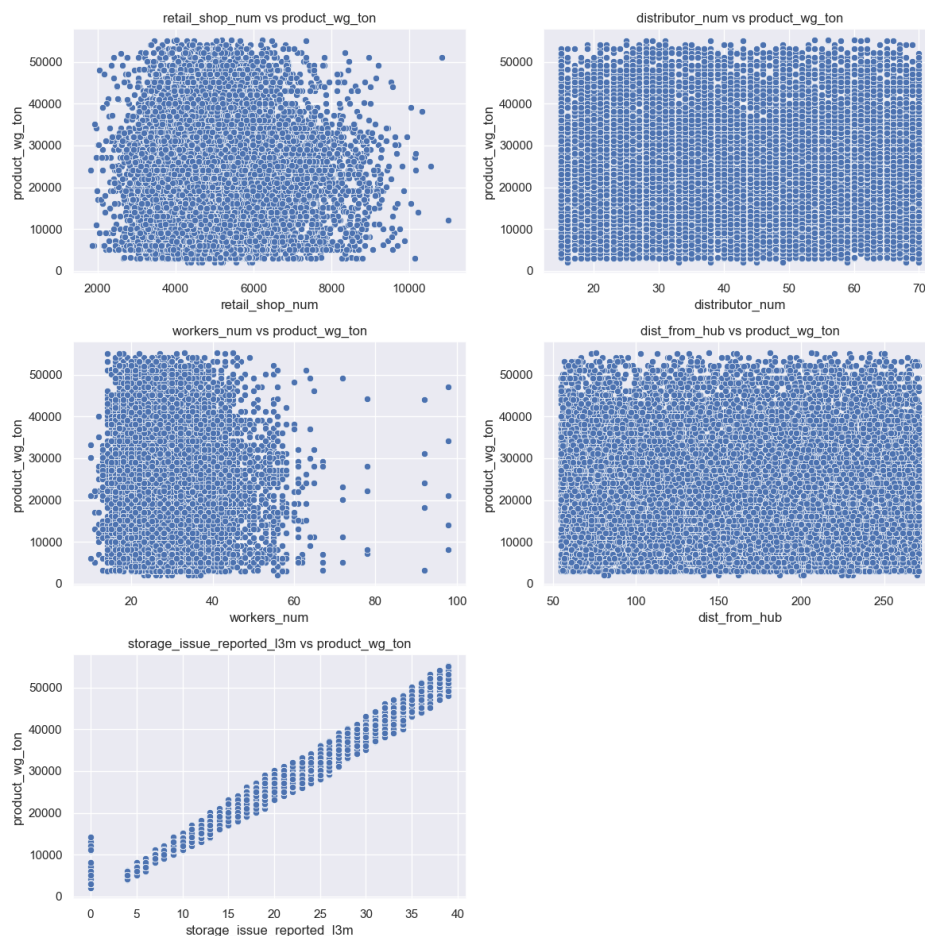


Fig 5. Numerical vs Target

1. retail_shop_num vs product_wg_ton

Observations: Most of the data points are clustered between 3000 to 8000 shops and 5000 to 50000 product weight. No strong trend is visible

Insights: retail shop number alone is not a strong predictor. Some regions may generate higher throughput despite fewer shops.

2. distributor_num vs product_wg_ton

Observations: The data points are high dispersed.

Insights: Distributor count has very weak influence on throughput.

3. workers_num vs product_wg_ton

Observations: Most of the warehouses work with 15-50 workers. Product weight remains highly variable across worker counts. Few high worker outliers exist but don't show higher throughput.

Insights: Workforce is alone is not correlated with shipment value, one of possible reasons could be workload imbalance.

4. dist_from_hub vs product_wg_ton

Observations: Data points are evenly distributed across all hub distances. No trend visible.

Insights: Distance from the hub does not affect the product throughput.

5. storage_issue_reported_l3m vs product_wg_ton

Observations: Strong linear positive trend observed. As storage issue increases, product weight increases.

Insights: High volume warehouses naturally experience more storage issues. They increase operational stress.

Categorical vs Target

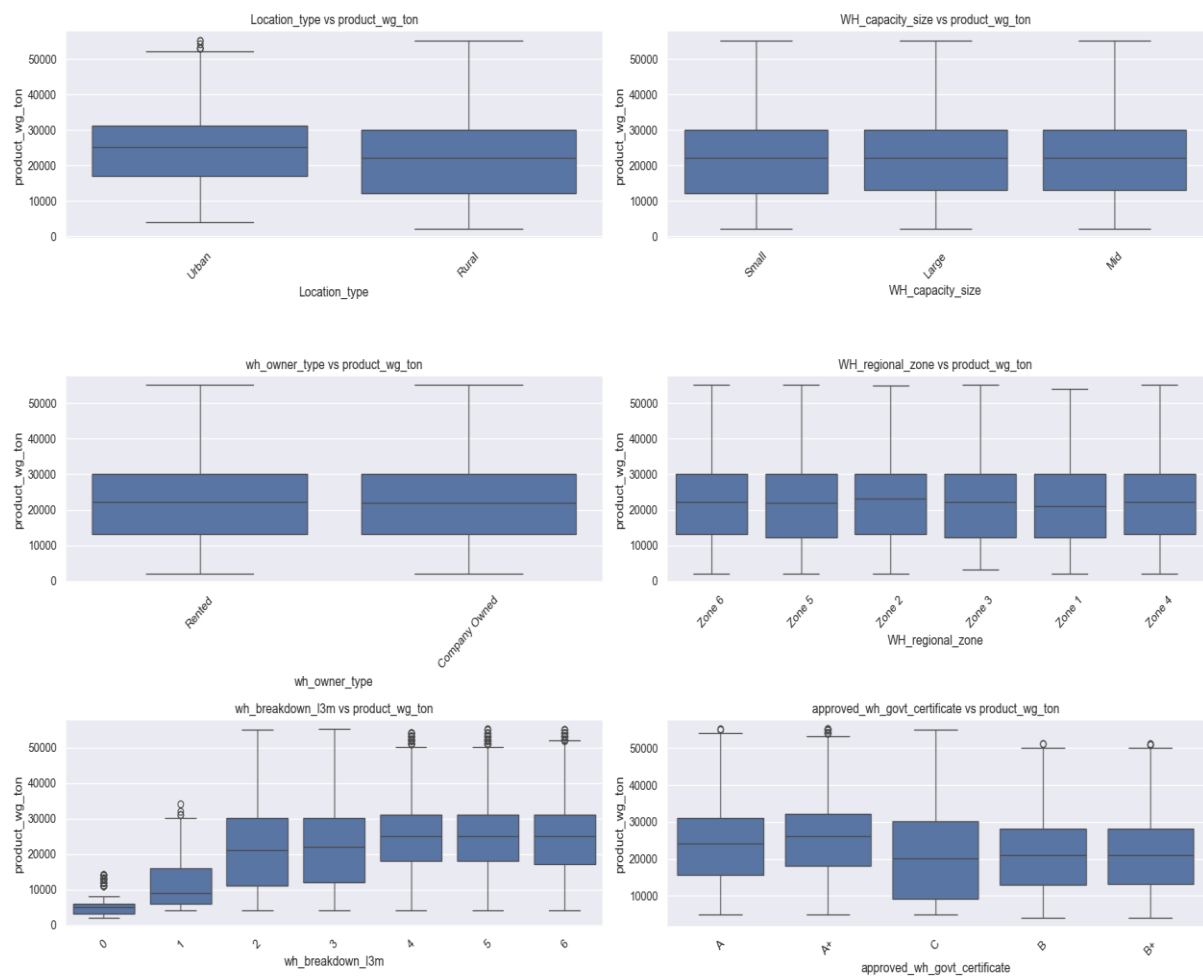


Fig 6. Categorical vs Target

1. Location_type vs product_weight_ton

Observations:

- Urban locations show a slightly higher median than rural.
- Rural shows a wider spread.
- Urban contains a few high value outliers.

Insights:

- Urban warehouses may handle more inventory due to better infrastructure, more demand, and better logistics connectivity.
- Rural operations may face fluctuations in demand and supply. They may require more planning.

2. WH_capacity_size vs product_weight_ton

Observations:

- The medians of all 3 sizes are similar.
- Small category has a wider spread compared to Mid and Large.

Insights:

- Warehouse size alone is not a strong predictor of product ton.

3. wh_owner_type vs product_wg_ton**Observations:**

- Both Rented and Company owner shows similar median and similar spread.

Insights:

- Company owned warehouses are more flexible in terms of handling strategic/high volume operations.

4. WH_regional_zone vs product_wg_ton**Observations:**

- All 6 regional zones show very similar distribution.
- Zone 2, 3 and 4 have a slightly higher median than the other regional zones.

Insights:

- Product movement appears geographically balanced across all 6 regional zones.
- No major imbalance exists.

5. wh_breakdown_l3m vs product_wg_ton**Observations:**

- Higher breakdown frequency corresponds to higher product weight.
- Warehouses with 4-6 breakdowns show significantly higher median product weight.

Insights:

- High volume warehouses experience more operational stress.
- Large throughput increases equipment which increases breakdown probability.

6. approved_wh_govt_certificate vs product_wg_ton**Observations:**

- All 5 categories show moderate difference.
- Category C appears to have lower throughput and larger variability.

Insights:

- Government certified warehouses may handle large scale operations and follow stricter protocols.
- Certifications may positively influence product throughput.

Binary columns

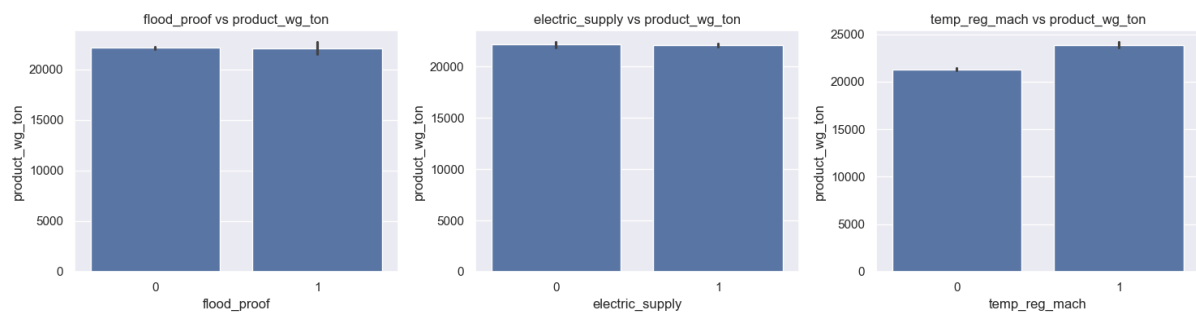


Fig 7. Binary columns

flood_proof vs product_wg_ton: Both categories 1 and 0 show almost identical average product weight.

electric_supply vs product_wg_ton: Both categories 1 and 0 show almost identical average product weight.

temp_reg_mach vs product_wg_ton: Warehouses with temperature regulating machines show higher average product weight than those without.

Insights:

- Flood proof capability and electric infrastructure do not significantly increase product movement.
- Temperature regulated facilities are associated with high product movement. It may indicate higher capacity warehouses, better infrastructure, good product handling, efficient inventory management.

2.5 Multivariate analysis

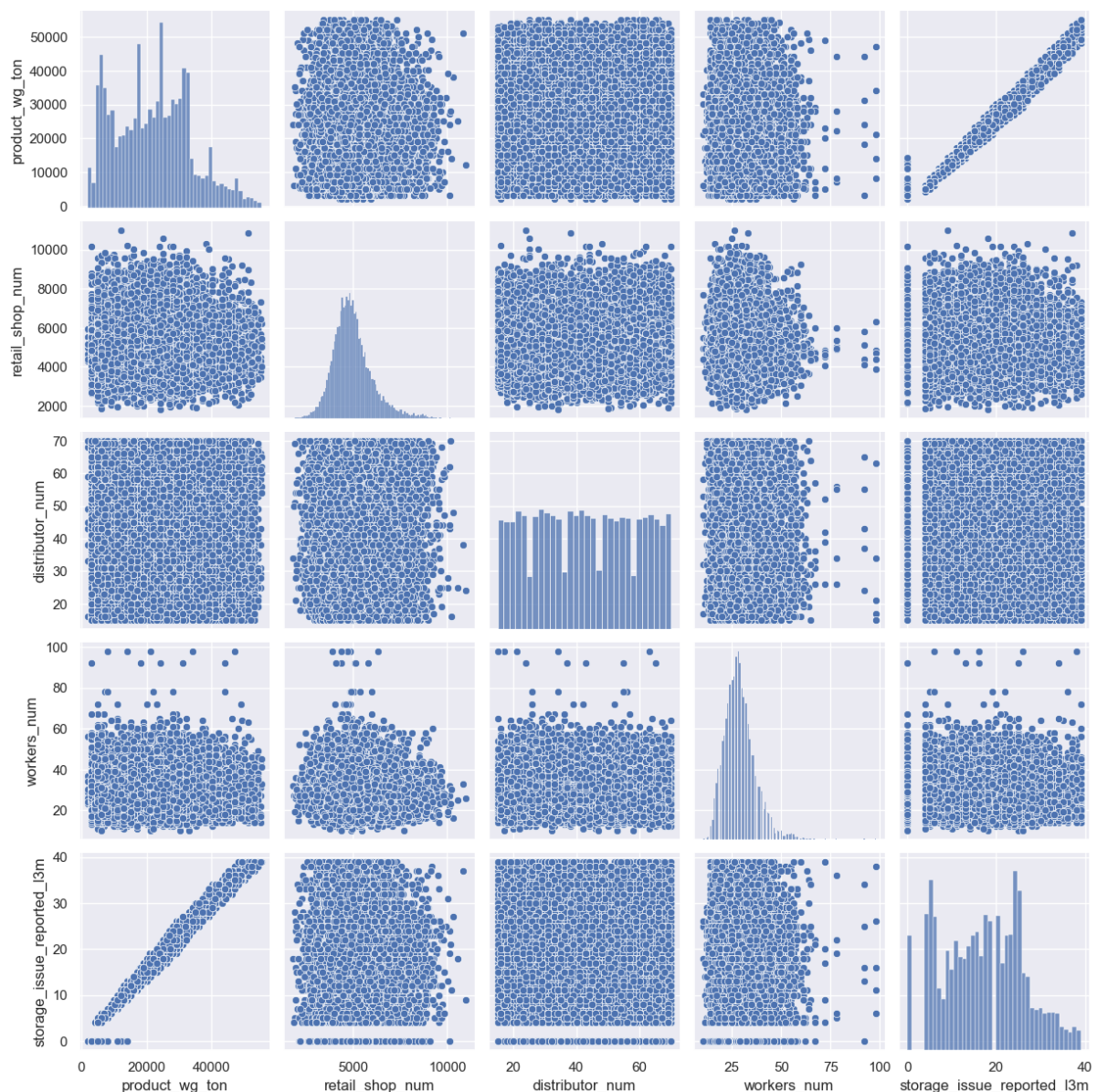


Fig 8. Pairplot

Observations:

- The pairplot indicates that several variables exhibit non-linear or weak relationships with the target variable, which may limit the performance of simple linear models.
- 'product_wg_ton' shows a very strong positive relationship with storage_issue_reported_l3m.
- Warehouses with larger product quantity tend to have more storage issues.
- 'retail_shop_num' does not show any linear relationship but moderate clustering patterns are visible.
- 'distributor_num' is evenly distributed and does not exhibit strong direct correlation with product movement.
- 'workers_num' appears moderately concentrated within a limited range, suggesting workforce allocation is relatively standardized across warehouses.

3. Data Preprocessing

3.1 Duplicate value check

There are no duplicates in the dataset.

3.2 Data cleaning

'Ware_house_ID' and 'WH_Manager_ID' are removed as they do not contribute any meaningful information.

3.3 Anomalous value check

Anomalous value check was performed on numerical and categorical datatype. There are no negative values found and binary variables are also consistent.

3.4 Missing value check

3 variables have missing values:

Feature	Missing Values	Treatment
Workers_num	990	Median imputation
wh_est_year	11881	Converted to Warehouse age
approved_wh_govt_certificate	908	Mode imputation

Table 2. Missing value treatment

3.5 Feature Engineering

Warehouse Age

The 'wh_est_year' column contained large number of null values. So instead of dropping the column, another column was created named 'wh_age'.

Steps performed were:

- 'wh_est_year' was converted into numerical format.
- 'wh_age' was calculated using

$$\text{Warehouse Age} = 2026 - \text{Warehouse establishment year}$$

- A binary missing indicator feature wh_age_missing was created to capture whether the original wh_est_year was missing.
- Missing values were imputed using median in 'wh_age' column and 'wh_est_year' was dropped.

Infrastructure Score

An additional column named 'infra_score' was created involving 3 features:

$$\text{Infrastructure Score} = \text{Electric Supply} + \text{Flood proof} + \text{Temperature Regulation Machine}.$$

The resulting score ranges from 0 to 3, where higher values indicate better warehouse infrastructure.

Since the information of these 3 features are captured in the new variable 'infra_score', we drop the 3 features.

3.6 Outlier Check and Treatment

Outliers were analysed using boxplot. Potential outliers were observed in:

- Retail_shop_num
- Workers_num
- Competitor_in_mkt

Since supply chain datasets naturally contain extreme values due to region demands and warehouse capacities, removing these records could lead to loss of information.

Therefore, IQR based capping method aka Winsorization was applied to limit extreme values. This approach helped reduce the impact of extreme observations on the regression model.

3.7 Class Imbalance handling

Since this is a regression-based problem, class imbalance handling techniques were not applicable.

3.8 Data Preparation for Modelling

Encoding of Categorical Variables: ML models cannot directly process categorical data. So we convert it into numerical format using One-hot Encoding.

The drop_first=True parameter was used to avoid multicollinearity.

3.9 Feature Scaling

Feature scaling was performed using StandardScaler technique. Scaling helps improve model stability and ensures that variables with larger numerical ranges do not dominate the learning process.

3.10 Data Leakage Handling

- Train test split was performed before feature scaling
- Unnecessary features were removed after feature engineering.
- The scaler was fitted on the training data and then applied to the test data.

4. Model Building - Baseline model

4.1 Metric of Choice

Since the target variable is a continuous variable, the problem was treated as regression problem. Multiple regression metrics were used to assess model performance including RMSE, MAE, R² Score, Adjusted R² Score and MAPE.

- **RMSE (Root Mean Squared Error):** measures average prediction error while giving higher importance to large errors.
- **MAE (Mean Absolute Error):** measures average absolute prediction error.
- **R² Score:** indicated the proportion of variance explained by the model.

- **Adjusted R² Score:** measures the variance explained by the model while accounting for the number of predictors used.
- **MAPE (Mean Absolute Percentage Error):** measures the average prediction error as percentage of actual values.

Among these, RMSE (Root Mean Square Error) was selected as the primary metric as it measures the average prediction error in the same unit as the target variable and it is widely used in supply chain predictions where large errors can significantly impact business operations.

4.2 Baseline Model Development

A **Linear Regression** model was developed as the baseline model.

The dataset was pre-processed before modelling: data cleaning, missing value check, feature engineering, outlier check and treatment.

The dataset was then split into 80:20 training and testing sets.

4.3 Model Performance Evaluation

Model 1: dataset with 'storage_issue_reported_l3m'

	Training performance	Testing performance
RMSE	1384.53	1336.95
MAE	1008.75	993.37
R² Score	0.98	0.98
Adj R² Score	0.98	0.98
MAPE	7.60	7.40

Table 3. LR (model 1)

Training performance:

- Low RMSE (1384.53) and MAE (1008.75) indicates low prediction error and that predictions are close to the actual values.
- Model explains 98.5% variance in training data.
- Average prediction error is approximately 7.6%.

Testing performance:

- Low RMSE (1336.95) and MAE (993.37) indicates low prediction error and that predictions are close to the actual values.
- Model explains 98.6% variance in testing data.
- Average prediction error is approximately 7.4%.

LR model demonstrates very strong predictive performance on both training and testing sets. The values of train and test metrics are very close, indicating good generalisation and maybe overfitting. The high performance may be influenced by 'storage_issue_reported_l3m' as it was highly correlated.

Since correlation between 'product_wg_ton' and 'storage_issue_reported_l3m' is 0.99, we can try another model with storage_issue_reported_l3m feature.

Model 2: dataset without 'storage_issue_reported_l3m'

	Training performance	Testing performance
RMSE	8409.48	8351.16
MAE	6621.44	6615.68
R² Score	0.47	0.47
Adj R² Score	0.47	0.47
MAPE	48.14	48.38

Table 4. LR (model 2)

Training performance:

- RMSE (8409.48) and MAE (6621.44) increased significantly.
- Now models explain only 47.6% variance in training data.
- Average prediction error is approximately 48%.

Testing performance:

- RMSE (8351.16) and MAE (6615.68) increased significantly.
- Now models explain only 47.8% variance in testing data.
- Average prediction error is approximately 48%.

Comparison of 2 models on testing set:

Metric	With feature	Without feature
RMSE	1336.95	8351.16
MAE	993.37	6615.68
R² Score	0.98	0.47
Adj R² Score	0.98	0.47
MAPE	7.40	48.38

Table 5. LR (model 1 vs model 2)

After removing 'storage_issue_reported_l3m' feature, model performance dropped significantly. This indicates that the feature had very strong relationship with the target variable. Although 47.9% is moderate, the model is considered more realistic and reliable for practical business applications. Model 2 can serve as a baseline model for future improvements using advanced machine learning algorithms.

5. Model Building – Advanced Models

Advanced models and tree-based regression models were developed to improve the prediction accuracy and capture complex non linear relationships in the data.

To avoid potential data leakage, the Model 2 i.e., the dataset without 'storage_issue_reported_l3m' feature is used.

5.1 Models Used

The following models were implemented:

1. Decision Tree Regressor
2. Bagging Regressor
3. Random Forest Regressor
4. AdaBoost Regressor
5. Gradient Boost Regressor
6. XGBoost Regressor

The performance Metrics used were

- RMSE
- MAE
- R^2 score
- Adjusted R^2 score
- MAPE

5.2 Model Performance Comparison

Training performance Comparison:

	Decision Tree	Bagging Regressor	Random Forest Regressor	AdaBoost Regressor	Gradient Boost	XGBoost
RMSE	0.0	3484.67	2966.51	8235.85	7700.22	5829.63
MAE	0.0	2466.50	2248.54	6575.14	5906.21	4410.73
R^2	1.0	0.91	0.93	0.49	0.56	0.74
Adj R^2	1.0	0.90	0.93	0.49	0.56	0.74
MAPE	0.0	15.81	14.69	48.63	38.89	29.22

Table 6. Training performance Comparison

Testing performance Comparison:

	Decision Tree	Bagging Regressor	Random Forest Regressor	AdaBoost Regressor	Gradient Boost	XGBoost
RMSE	11008.27	8270.28	7813.64	8162.60	7695.54	8043.84
MAE	8217.12	6277.49	5972.88	6526.45	5924.73	6207.17
R^2	0.09	0.48	0.54	0.50	0.55	0.51
Adj R^2	0.08	0.48	0.54	0.49	0.55	0.51
MAPE	50.07	40.94	39.34	48.43	39.17	40.83

Table 7. Testing performance Comparison

Observations:

1. Decision Tree Regressor:

- The training model shows severe overfitting. The tree has memorised the training data perfectly.
- Training model has a perfect score with RMSE = 0 and $R^2=1$
- The model performs very poorly on the unseen data with RMSE = 11008.27 and $R^2=0.09$
- MAPE of 50.08% indicates large prediction error on the testing data.
- Decision Tree model fails to generalise.

2. Bagging Regressor

- Training model shows strong performance with RMSE = 3484.67 and $R^2=0.9$.
- Testing model shows moderate performance with RMSE = 8270.28 and $R^2=0.48$.
- MAPE of 40.95% indicates moderate prediction accuracy.
- Bagging Regressor model reduced overfitting compared to Decision Tree.

3. Random Forest Regressor

- Training model shows much better performance than Bagging with RMSE = 2966.51 and $R^2=0.93$.
- The training performance shows the lowest error so far.
- Testing model still shows moderate performance with RMSE = 7813.66 and $R^2=0.54$ but better than Bagging Regressor.
- MAPE of testing shows 39.34% indicating low prediction error.
- Random Forest shows strong predictive capability and better generalisation.

4. AdaBoost Regressor

- Moderate training performance with RMSE = 8235.85 and $R^2=0.49$
- The training models shows high training errors.
- The testing performance shows similar values to training with RMSE = 8162.60 and $R^2=0.50$.
- MAPE of testing is similar to training with 48.43%.
- AdaBoost Regressor shows stable performance but weaker compared to other models.

5. Gradient Boost Regressor

- Moderate training performance with RMSE = 7700.22 and $R^2=0.56$.
- Moderate training errors with MAPE of 38.89%.
- The testing values shows similar performance to training values.
- Testing performance has achieved the best R^2 (0.55), lowest RMSE (7695.54) and lowest MAE (5924.73).
- Gradient Boosting provided the best balance between bias and variance and demonstrated strong generalisation.

6. XGBoost Regressor

- The training set shows a strong performance with RMSE = 5829.63 and $R^2=0.74$.
- It has better learning capability than AdaBoost and Gradient Boosting regressor.
- Testing set shows moderate performance with RMSE = 8043.84 and $R^2=0.51$.
- MAPE of 40.83% shows moderate prediction accuracy.
- XGBoost performed well overall but it slightly underperformed compared to Gradient boosting and Random Forest.

6. Model Performance Improvement using Hyperparameter Tuning

GridSearchCV was used to identify the optimal combination of hyperparameters for the selected models.

6.1 Top 3 model for Tuning

Three high performing models were selected for hyperparameter tuning.

1. Random Forest Regressor

- Achieved strong training and testing performance.
- Produced low RMSE and MAE values compared to other models.
- Ensemble averaging helps reduce overfitting and improve model stability.
- Hyperparameter tuning can further optimize tree depth, number of estimators, and sample splitting.

2. Gradient Boost

- Achieved best testing performance among all models.
- The model shows strong generalisation capability with balanced train-test metrics.
- Boosting methods iteratively improve weak learners.
- Tuning the learning rate, depth and number of estimators can significantly improve the prediction accuracy.

3. XGBoost

- The model delivered a strong predictive performance and effectively captured non-linear relationships.
- XGBoost is an advanced boosting algorithm known for regularisation.
- The model has many tuning parameters that can substantially enhanced performance and reduce overfitting.

6.2 Performance of Tuned Models

	Random Forest		Gradient Boost		XGBoost	
	Train	Test	Train	Test	Train	Test
RMSE	7521.94	7693.57	7542.49	7668.57	6353.32	7885.62
MAE	5716.17	5882.68	5758.77	5886.29	4822.72	6071.41
R^2	0.58	0.55	0.57	0.56	0.70	0.53
Adj R^2	0.58	0.55	0.57	0.55	0.70	0.53
MAPE (%)	37.61	38.64	38.01	39.02	31.80	39.95

Table 8. Tuned model performance

1. Tuned Random Forest Regressor

- The training and testing metrics are very similar, suggesting that overfitting has been reduced after tuning.
- The test R^2 of 0.56 indicates that it explains approximately 56% of the variation in product weight.
- The RMSE (7693.57) and MAE (5882.68) indicates moderate prediction errors.
- MAPE of 38.64% shows moderate prediction accuracy.

2. Tuned Gradient Boosting Regressor

- Training and testing performance values are closely aligned showing a good balance between bias and variance.
- This model achieved the best test R^2 of 0.56 among the tuned models.
- Achieved the lowest RMSE (7668.57) and MAE (5886.26) indicating best predictive accuracy.

3. Tuned XGBoost Regressor

- The model achieved the best training performance with $R^2 = 0.70$.
- However, the test performance dropped to 0.53, indicating some overfitting.
- This model produced the highest RMSE (788.62) and MAE (6071.41) among other tuned models.
- The performance on unseen data remained weaker than Random Forest and Gradient boosting.

7. Model Performance Comparison and Final Model Selection

7.1 Comparison of Tuned Models (Test data)

	Tuned Random Forest	Tuned Gradient Boosting	Tuned XGBoost
RMSE	7693.57	7668.57	7885.62
MAE	5882.68	5886.29	6071.41
R^2	0.55	0.56	0.53
Adj R^2	0.55	0.55	0.53
MAPE (%)	38.64	39.02	39.95

Table 9. Comparison of tuned models

7.2 Final model Selection

Based on the above testing performance, the **Tuned Gradient Boosting Regressor** was selected as the final model.

This model achieved:

- Highest test R^2 score
- Lowest test RMSE

- Stable performance across train and test sets and
- Strong generalisation capability.

The final Gradient Boosting model was able to explain approximately 56% of the variance in the warehouse product movement while maintaining stable prediction performance on unseen data. The model effectively captured non-linear relationships and operational interactions among supply chain variables, making it suitable for predictive decision-making in warehouse and logistics management.

8. Actionable Insights and Recommendations

8.1 Key Insights

- The variable 'storage_issue_reported_l3m' showed a very strong relationship with the target variable 'product_wg_ton'. Warehouses reporting higher storage issues had higher product movement. This feature was later identified as a potential data leakage variable and removed from the baseline model.
- Warehouses with better infrastructure conditions like electric supply, flood proof facilities, temperature regulation systems showed better warehouse functioning and product handling.
- Warehouse capacity and workforce affected the product shipment. Larger warehouses with more workforce and greater distributor coverage handled larger product volumes.
- Geographical factors influenced the warehouse performance. Warehouses located farther from distributor hubs experienced lower operational efficiency due to transportation and delivery challenges.

8.2 Actionable Business Recommendations

- The company should invest in temperature regulation systems, backup power supply and flood proof mechanisms to improve the efficiency and reduce the disruptions.
- According to the numbers, the workforce doesn't really affect the product movement. So, focus on productivity improvement by implementing training, using technology and automation.
- Increase storage capacity to accommodate growing product volumes and reduce storage related issues.
- Consider expanding warehouse capacity in high volume regions to reduce delivery times and improve customer service levels
- Lower certified warehouses can adopt best practices from highly certified warehouses to improve the quality standards, and overall performance.

APPENDIX

Appendix A: Data Preprocessing & Feature Engineering

A1. Missing Value Check and Treatment

```
df.isnull().sum()
```

```
Location_type      0
WH_capacity_size    0
zone               0
WH_regional_zone    0
num_refill_req_13m  0
transport_issue_11y 0
Competitor_in_mkt   0
retail_shop_num     0
wh_owner_type       0
distributor_num     0
flood_impacted      0
flood_proof         0
electric_supply     0
dist_from_hub       0
workers_num        990
wh_est_year        11881
storage_issue_reported_13m 0
temp_reg_mach       0
approved_wh_govt_certificate 908
wh_breakdown_13m    0
govt_check_13m      0
product_wg_ton      0
dtype: int64
```

Fig 9. A1 Missing Value Check

```
# 'workers_num' - apply median
df['workers_num'] = pd.to_numeric(df['workers_num'])
df['workers_num'] = df['workers_num'].fillna(df['workers_num'].median())

# 'approved_wh_govt_certificate' - apply mode
df['approved_wh_govt_certificate'] =
df['approved_wh_govt_certificate'].fillna(df['approved_wh_govt_certificate']
    '.mode()[0])
```

Missing values in 'wh_est_year' were handled indirectly through feature engineering by creating 'wh_age' and imputing missing age values using median imputation.

A.2 Feature Engineering

Warehouse Age (wh_age)

```
#convert into numeric
df['wh_est_year'] = pd.to_numeric(df['wh_est_year'])

#Warehouse age
df['wh_age'] = 2026 - df['wh_est_year']

#Missing indicator
df['wh_age_missing'] = df['wh_age'].isna().astype(int)
```



```
#impute missing values using median
df['wh_age'] = df['wh_age'].fillna(df['wh_age'].median())
```

Infrastructure score

```
df['infra_score'] = ( df['electric_supply'] + df['flood_proof'] +
df['temp_reg_mach'])
df['infra_score']
```

The infrastructure score ranges from 0 to 3, where higher values indicate better warehouse infrastructure availability.

A.3 Outlier check and treatment

```
# outlier detection using boxplot
num_cols = df.select_dtypes(include=np.number).columns.tolist()
```

```
plt.figure(figsize=(15, 9))
```

```
for i, variable in enumerate(num_cols):
    plt.subplot(4,5, i + 1)
    sns.boxplot(data=df, x=variable)
    plt.tight_layout(pad=2)
```

```
plt.show()
```

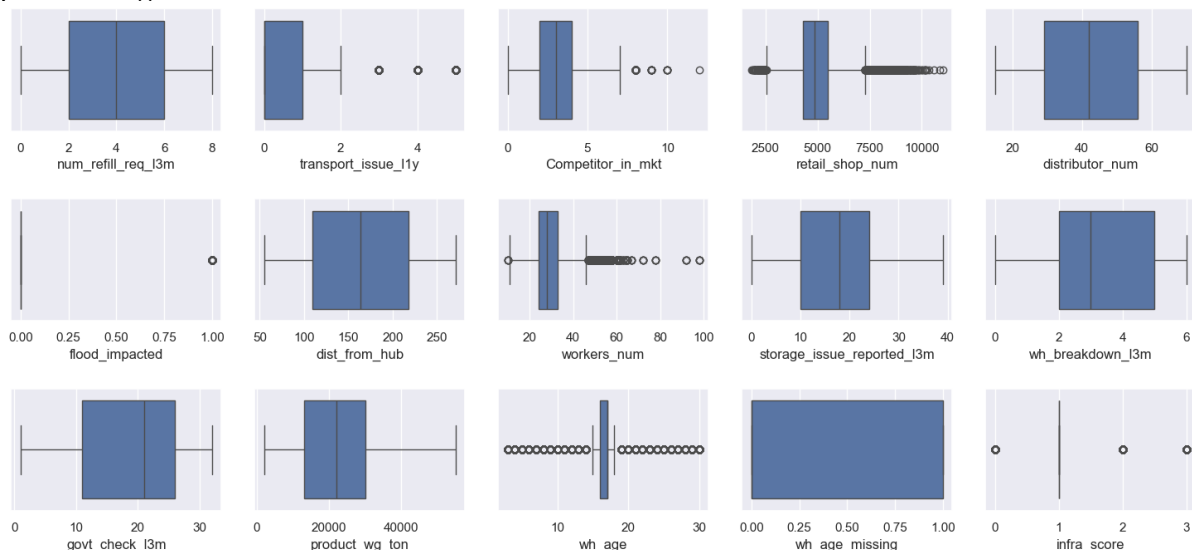


Fig 10. A3 Outlier detection

```
#Outlier treatment - Capping
outlier_cols = ['retail_shop_num', 'workers_num', 'Competitor_in_mkt',
'product_wg_ton']
```

```
for col in outlier_cols:
    Q1 = df[col].quantile(0.25)
```

```

Q3 = df[col].quantile(0.75)
IQR = Q3-Q1

lower = Q1-1.5*IQR
upper = Q3+1.5*IQR

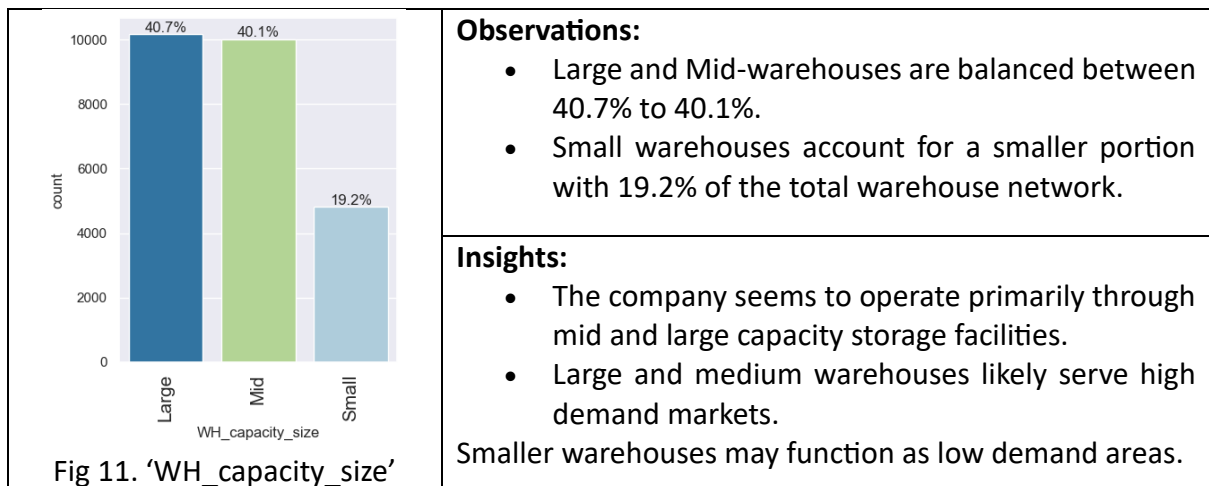
df[col] = df[col].clip(lower,upper)

```

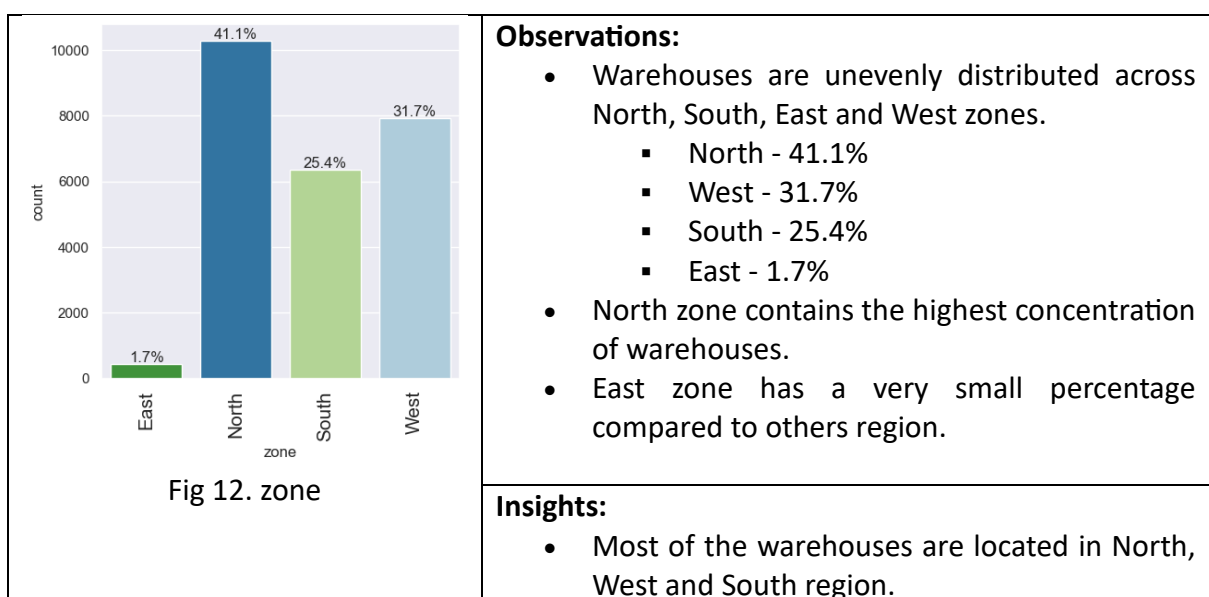
Outlier treatment was performed using the IQR capping method to reduce the influence of extreme values.

Appendix B: Detailed EDA – Univariate plots

1. WH_capacity_size

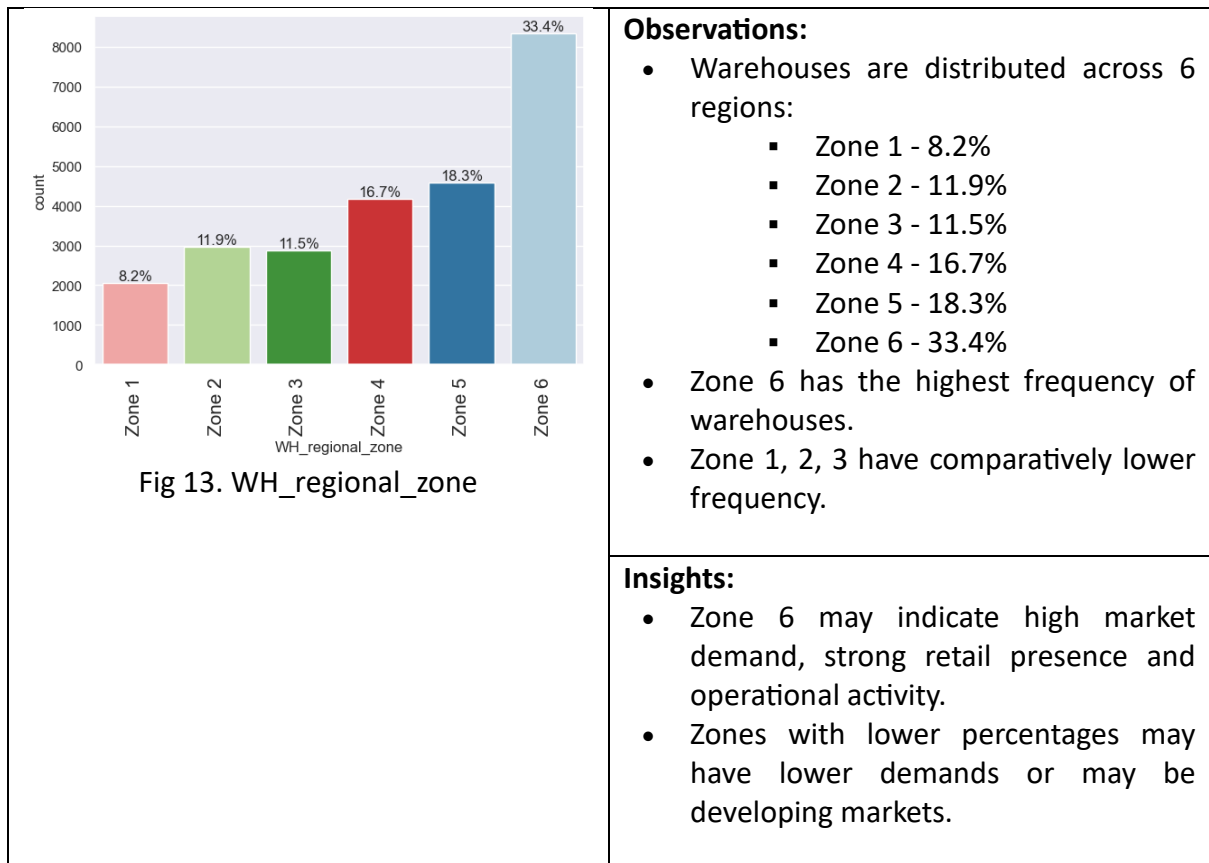


2. zone

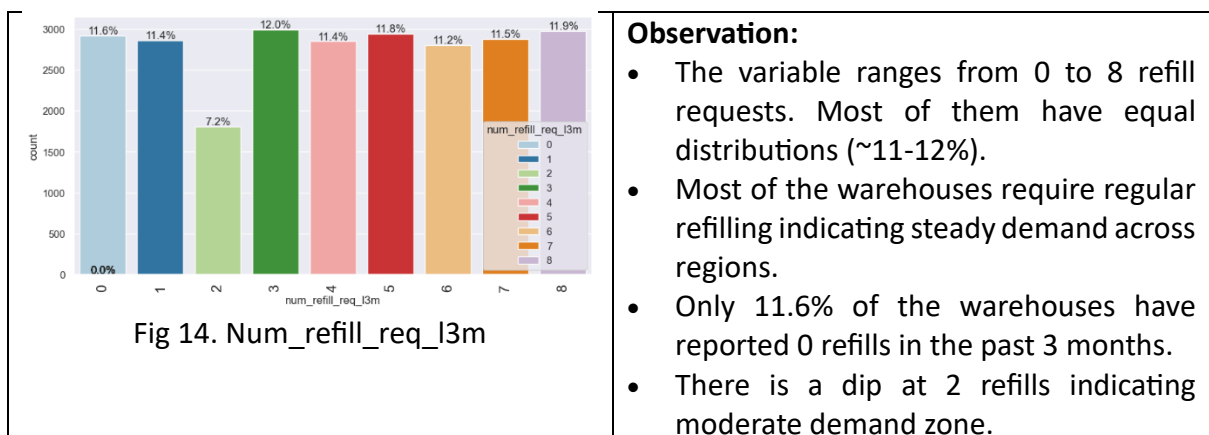


	• The high concentration in these 3 zones may indicate large market demand and high retail density. • The lower representation in East zone may indicate smaller customer base, limited operations, etc.
--	---

3. WH_regional_zone



4. num_refill_req_l3m



	Warehouses with 5-8 refills may indicate high demand zones and faster moving inventory hubs.
--	--

5. transport_issue_l1y

transport_issue_l1y	count	percentage
0	14000	60.9%
1	4500	18.6%
2	2200	8.8%
3	1800	7.3%
4	600	3.1%
5	300	1.4%

Fig 15. Transport_issue_l1y

Observations:

- The distribution is highly right skewed with most warehouses experiencing minimal transportation issues.
- Around 70% of the warehouses have reported 0 transport issues in the last year.
- 18.6% have experienced 1 transport issue.
- Very few warehouses have experienced more than 3 issues
 - 3 issues - 7.3%
 - 4 issues - 3.1%
 - 5 issues - 1.4%

Insights:

- Most warehouses seem to have efficient transportation system. A smaller number of warehouses faces regular issues indication poor road infrastructure, remote areas which may lead to delays in deliveries, increase in operational costs. They can have different routes or backup transportations.

6. Competitor_in_mkt

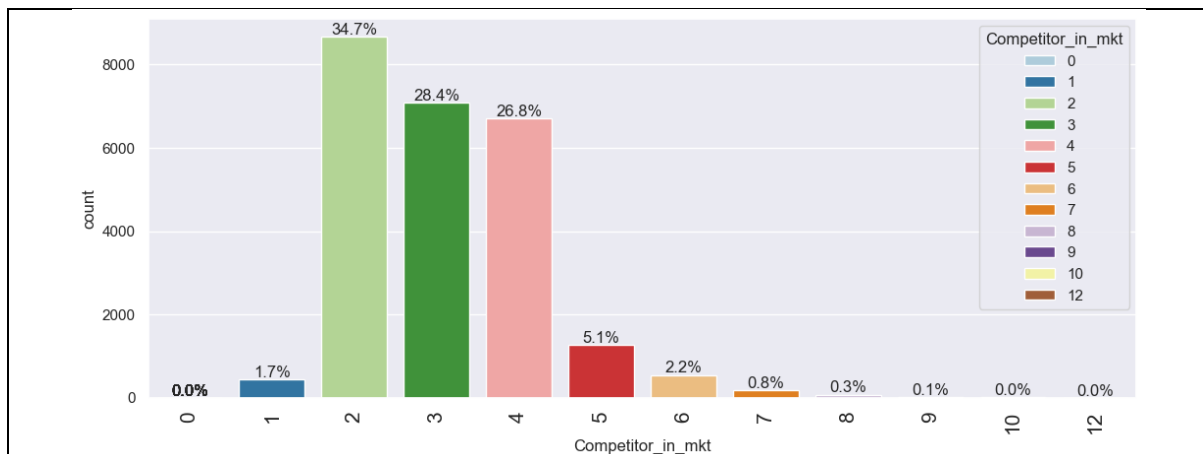


Fig 16. Competitor_in_mkt

Observations:

- Most of the markets have moderate level of competitors.
 - 2 competitors - 34.7%
 - 3 competitors - 28.4%
 - 4 competitors - 26.8%
- Around 1.7% have 1 and below competitors.
- Very few regions have extremely high competitors (6+)

Insights:

- 2-4 competitors suggest a healthy market demand and an active FMCG competition.
- Markets with more competitors may require more strong marketing & better pricing strategies.

7. retail_shop_num

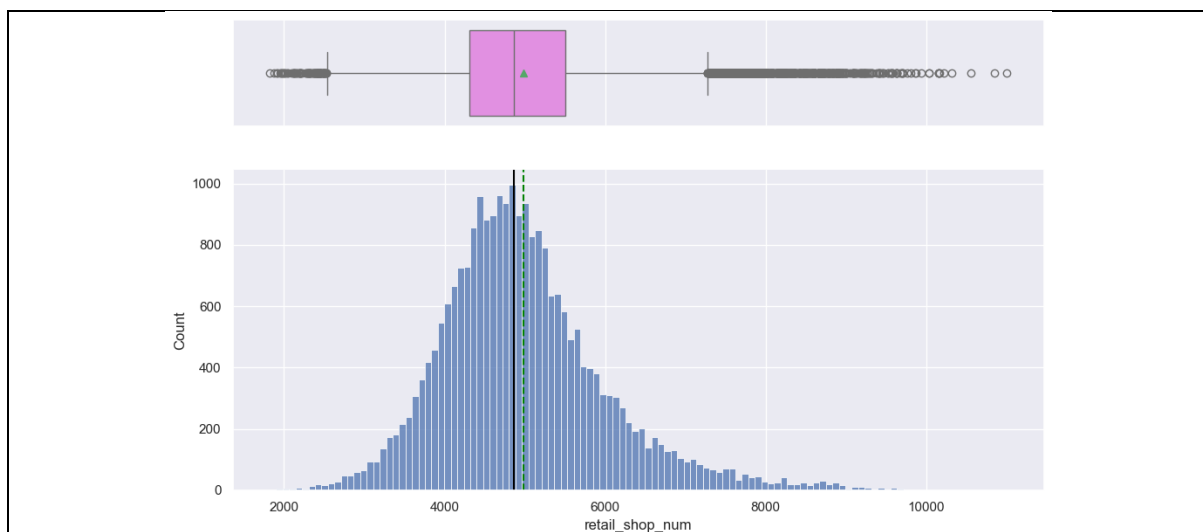


Fig 17. Retail_shop_num

Observations:

- The distribution is slightly skewed to the right.
- Most warehouses have 4000-5500 retail shops.

- Few outliers are seen below 3000 and high value outlier are present exceeding 10000.

The mean and median are almost close, indicating moderate skewness.

Insights:

- Regions with large number of retail shops may indicate densely populated areas with high demand. They may require higher inventory allocation, faster replenishment cycles and stronger distributor networks.
- Regions with smaller retails may indicate rural or low demand areas.
- The wide variation suggests that the company operates in both urban and rural areas.

8. wh_owner_type

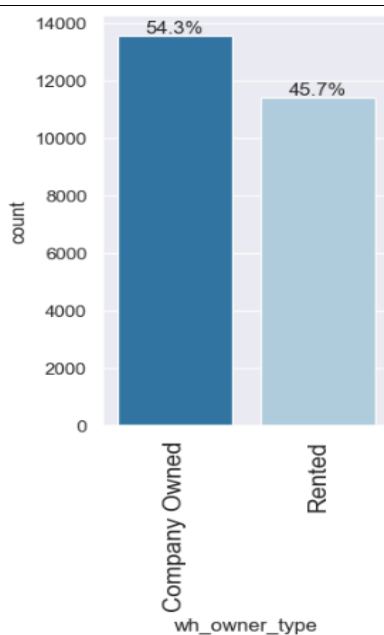


Fig 18. Wh_owner_type

Observations:

- Warehouses are almost evenly divided between Company Owned and Rented:
 - Company Owned – 54.3%
 - Rented – 45.7%
- Company owned warehouses is slightly more than Rented facilities.

Insights:

- The distribution indicates a balanced mix of owned and leased facilities.
- Company owned warehouses may have better operational control and better customization.
- Rented warehouses may help the company to reduce initial infrastructure investment.

9. distributor_num

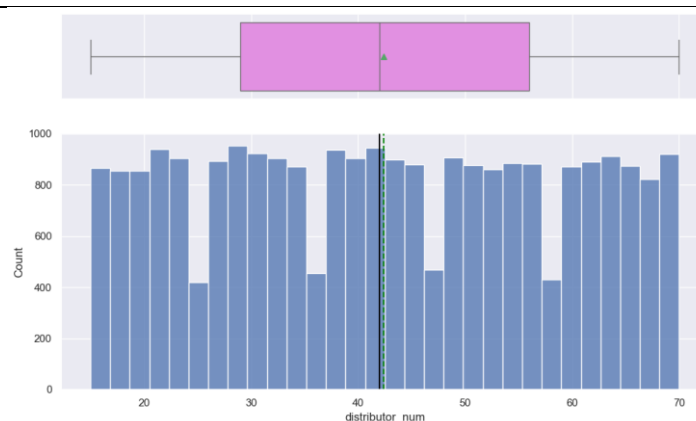


Fig 19. Distributor_num

Observations:

- The distribution seems uniform across the range 15-70.
- The mean and median are almost identical indicating a balanced distribution.
- No outliers or major skewness present.

Insights:

- The company maintains a fair number of distributors working between the warehouse and retail shops.
- Higher distributor count may support wider market coverage, reduce delays etc.
- Lower distributor count may indicate smaller market region.
- Balanced distributor suggests a stable logistics planning and standardized supply chain operations.

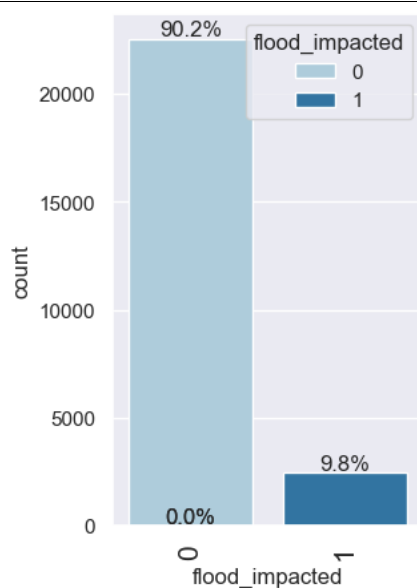
10. flood_impacted

Fig 20. Flood_impacted

Observations:

- 9.8% of the warehouses are located in the flood impacted region.
- 90.2% of the warehouses are not flood impacted.

Insights:

- Majority of the warehouse operates in a low-risk regions.
- A small set of warehouses are vulnerable to inventory damage, delays and disruption in workflow. This may lead to increase in operational cost, logistics uncertainty.
- These warehouses may require additional stocks, a recovery plan, infrastructure reinforcement.

11. flood_proof

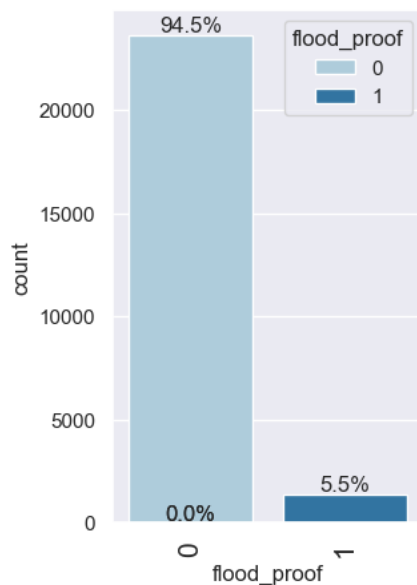


Fig 21. Flood_proof

Observations:

- 94.5% of the warehouses are not flood proof. These may be non-flood-impacted zones.
- The other 5.5% are flood proof. These may be the ones that are in the flood-impacted zones.

Insights:

- We see 9.8% of the warehouses are located in the flood impacted region. But only 5.5% are flood proof.
- Limited adoption of flood-proof infrastructure could suggest non risk zones, cost saving decisions.
- Warehouses with flood proof infrastructure could be located in high-risk zones, designed for safer inventory storage.

12. electric_supply

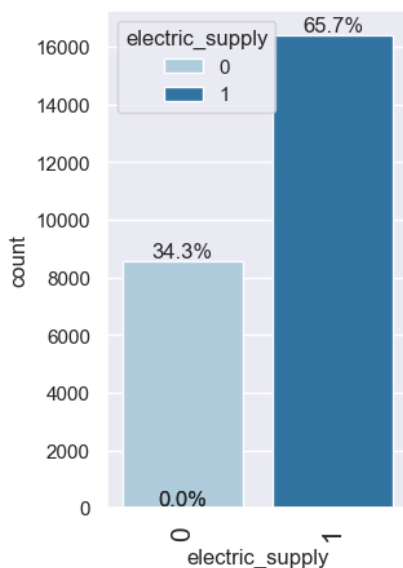


Fig 22. Electric_supply

Observations:

- 65.7% of the warehouses have electric backup supply.
- 34.3% of the warehouses lack electric backup supply.

Insights:

- Majority of the warehouses are equipped with power support systems where as a significant portion still operates with a backup power supply.
- Warehouses with electric supply are likely to experience smooth operations. Its important for temperature regulated storage, maintaining the product quality during power outages.
- Warehouses lacking backup power may face operational interruptions, inventory spoilage risks, reduce in operational efficiency.

13. dist_from_hub

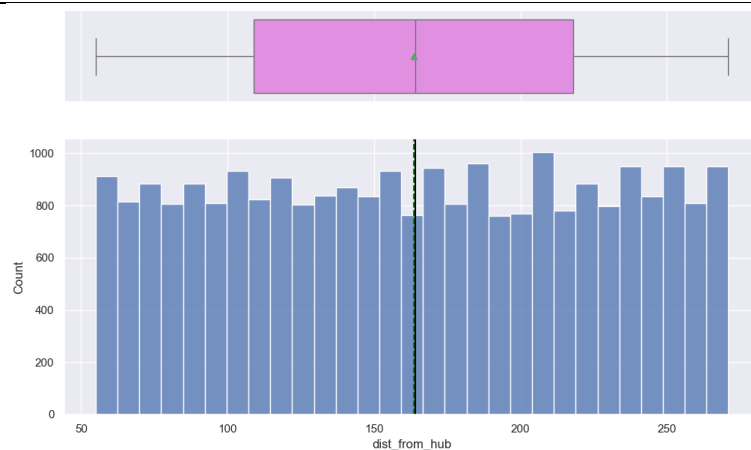


Fig 23. Dist_from_hub

Observations:

- The distribution is uniform across the range 55 to 270.
- The histogram shows no skewness or clustering.
- The boxplot shows no outliers.
- Mean and median are almost equal indicating a balanced distribution.

Insights:

- Warehouses are evenly distributed at both short and long distances from the hub. It seems like the company maintains a well-planned logistics infrastructure.

14. workers_num

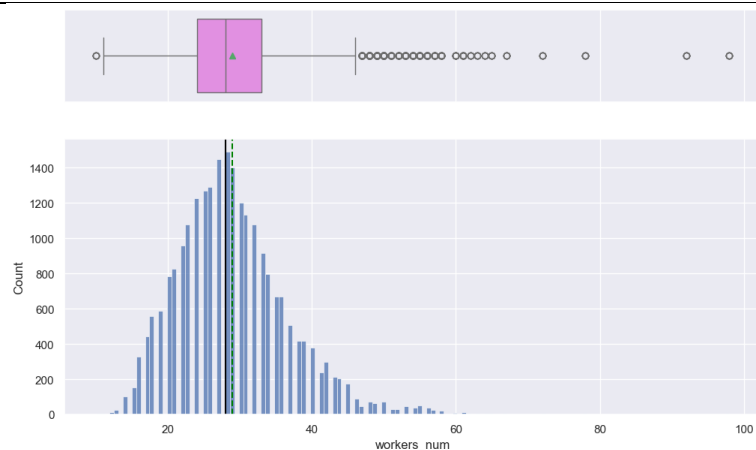


Fig 24. Workers_num

Observations:

- The distribution is right skewed.
- Most warehouses employ approx. 20 to 25 workers.
- Many warehouses have very large workforce size, extending up to nearly 100 workers.
- There are few high value outliers.

Insights:

- Many warehouses tend to operate with moderate workforce.
- Warehouses with high workers may indicate high demand centres, large capacity warehouses. They support faster inventory movement, increase in operational efficiency.
- Smaller workforce may indicate low demand regions and smaller storage facilities.

15. wh_est_year

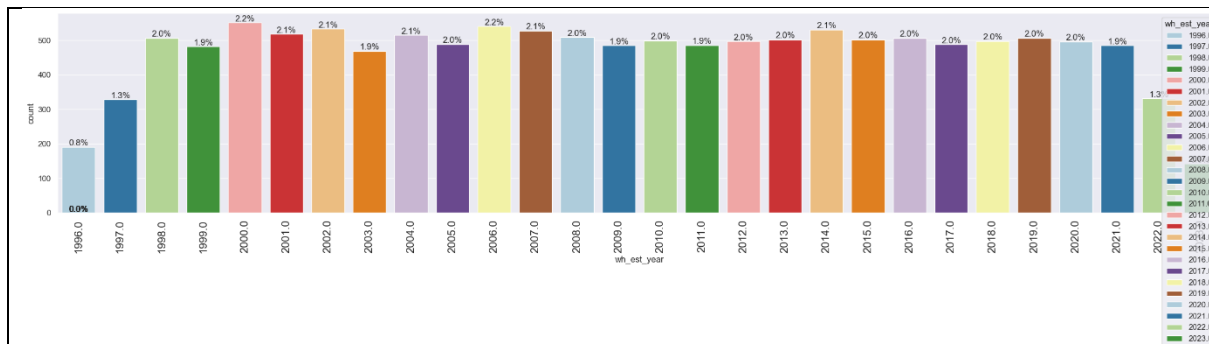


Fig 25. Wh_est_year

Observations:

- Warehouses were established between 1996 to 2022.
- The distribution uniform across the years.
- Most of the establishment years contribute 1.7% to 2.2% of the total warehouses.
- Early years like 1996-1997 shows very few warehouses.

Insights:

- The company seem to have expanded warehouse infrastructure steadily over time.
- Older warehouses may represent regions with long term demand, mature distribution centres, established operational hubs.
- Newly established warehouses may indicate market expansion, scaling of supply chain operations and penetrating into long term demands.

The balanced establishment distribution suggests a continuous operational growth of the company.

16. storage_issue_reported_l3m

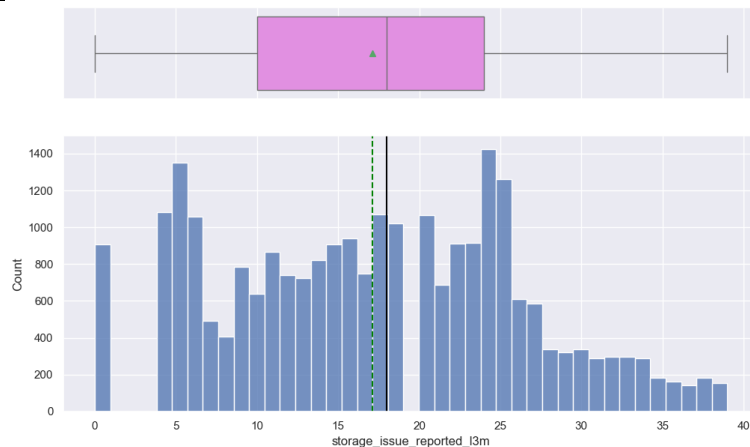


Fig 26. Storage_issue_reported_l3m

Observations:

- The distribution is almost uniform. The values ranges from 0 to 39.
- Most of the warehouses have reported storage issues between 10 and 25 times in the last 3 months.
- The mean and median are relatively close, indicating moderate skewness.
- The boxplot shows no outliers and a wide spread in storage issue frequency.

Insights:

- Warehouses with more issue count may indicate overcrowded storage facilities, poor storage practices, limited infrastructure that might lead to wastage of products, increase in operational costs.

Warehouses with lower issues likely have better storage facilities, stable inventory turnover.

17. temp_reg_mach

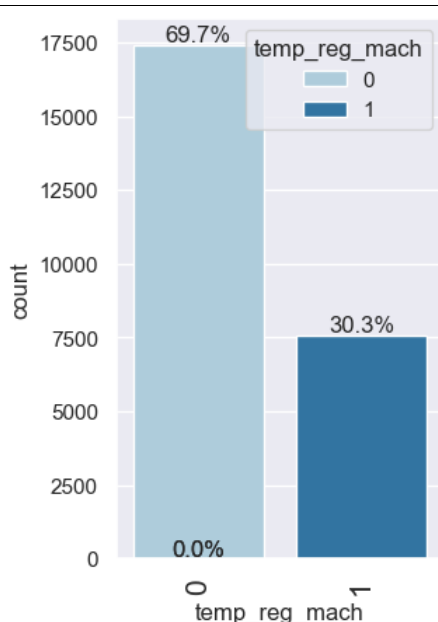


Fig 27. Temp_reg_mach

Observations:

- Around 69.7% of the warehouses do not have temperature regulating machines.

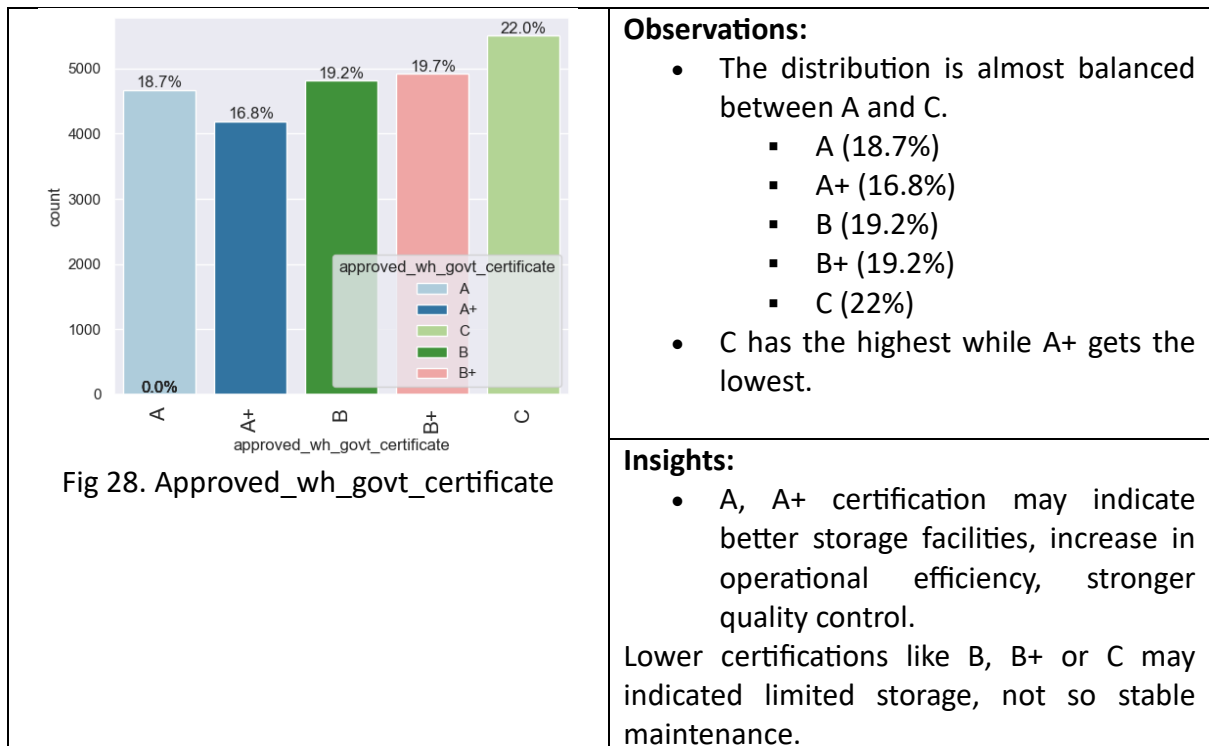
Only 30.3% of the warehouses are equipped with temperature regulating machines.

Insights:

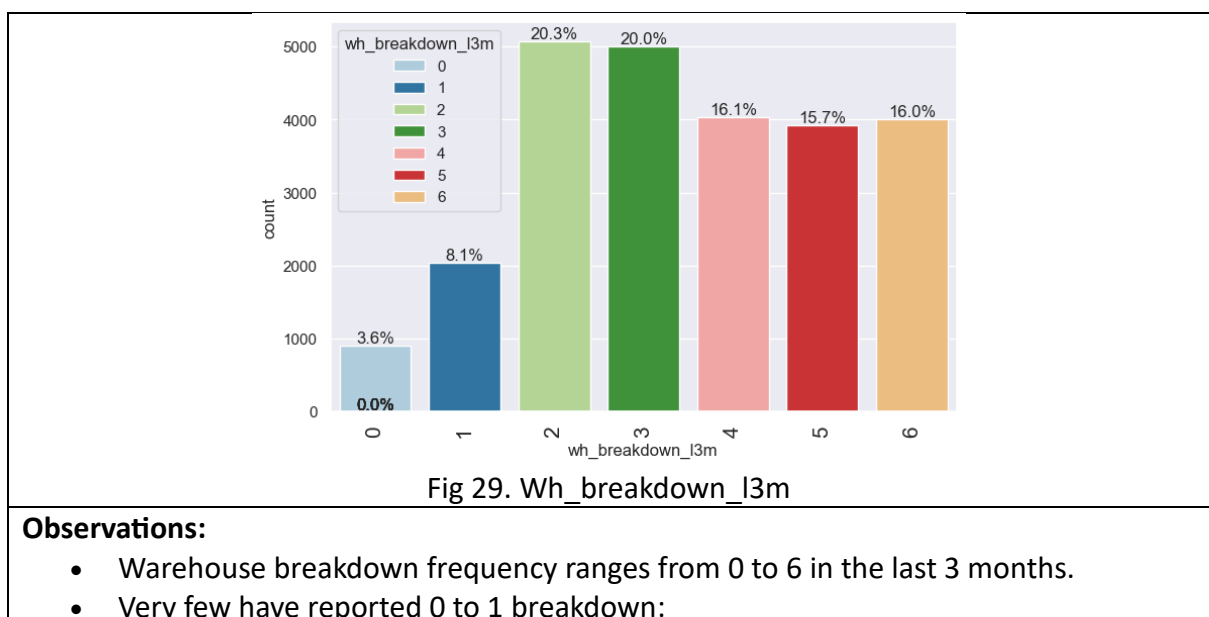
- Majority of the warehouses rely on standard storage conditions without temperature regulating machines.
- Warehouses with temperature regulating machines likely have better preservation of products, better storage quality with reduced the spoilage risk. These warehouses may be premium infrastructure facilities or high-volume warehouses.

	Warehouses with temperature regulating machines may indicate cost consideration towards warehouses that doesn't require advance storage facilities.
--	---

18. approved_wh_govt_certificate



19. wh_breakdown_l3m



- 0 breakdown – 3.6%
- 1 breakdown – 8.1%
- More than 15% of the warehouses have experienced 2 to 6 breakdowns.

Insights:

- A large portion of warehouses face recurring operational breakdowns.
- it may be due to worker strike, flood, electric failure which can lead to delay in deliveries, reduce in efficiency and face challenges in terms of inventory handling. These warehouses may require infrastructure maintenance, operational monitoring, etc.

20. govt_check_l3m

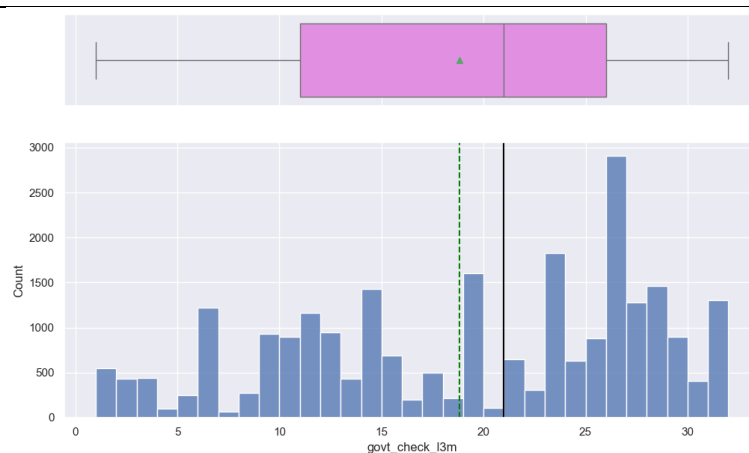


Fig 30. Govt_check_l3m

Observations:

- The distribution ranges from 1 to 32 checks in the last 3 months.
- The median is 21 and mean is 18, indicating slight left skewness.
- Most warehouses fall with 10-26 checks.
- No outliers are present.

Insights:

- The histogram shows multiple peaks, suggesting uneven inspections across warehouses.
- Variation in inspection frequency may depend on warehouse size and operational activity.

Appendix C: Comparison of tuned and untuned models

Training performance

	Random Forest		Gradient Boosting		XGBoost	
	Untuned	Tuned	Untuned	Tuned	Untuned	Tuned
RMSE	2966.51	7521.94	7700.22	7542.49	5829.63	6353.32
MAE	2248.54	5716.17	5906.21	5758.77	4410.73	4822.72
R ²	0.93	0.58	0.56	0.57	0.74	0.70
Adj R ²	0.93	0.58	0.56	0.57	0.74	0.70
MAPE	14.69	37.61	38.89	38.01	29.22	31.80

Table 10. Comparison of tuned and untuned models (training)

Testing performance

	Random Forest		Gradient Boosting		XGBoost	
	Untuned	Tuned	Untuned	Tuned	Untuned	Tuned
RMSE	7813.64	7693.57	7695.54	7668.57	8043.84	7885.62
MAE	5972.88	5882.68	5924.73	5886.29	6207.17	6071.41
R ²	0.54	0.55	0.55	0.56	0.51	0.53
Adj R ²	0.54	0.55	0.55	0.55	0.51	0.53
MAPE	39.34	38.64	39.17	39.02	40.83	39.95

Table 11. Comparison of tuned and untuned models (testing)

Appendix D: Observed vs Predicted Plot

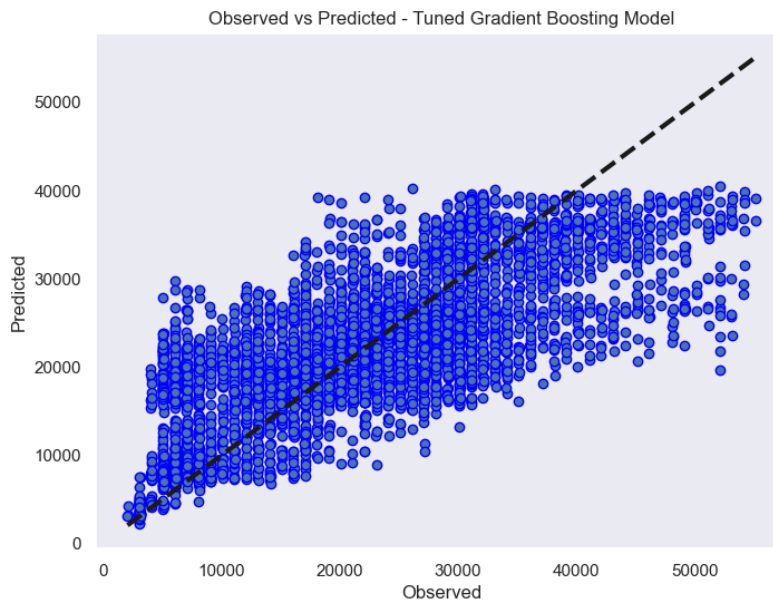


Fig 31. Observed vs Predicted plot

There is a clear positive trend between the observed and predicted values. Gradient Boosting model is able to capture the relationship between predictor variables and product weight.